CLASS 9 MATHEMATICS • CHAPTER 8

Predicting What Comes Next: Exploring Sequences and Progressions

Exhaustive point-by-point presentation covering every concept, explicit rule, recursive relation, arithmetic progression, geometric progression, and fractal directly from the textbook.

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What is a Sequence? Order, Terms & Infinity

FORMAL DEFINITION: SEQUENCE & TERM

A **sequence** is an ordered list of numbers or objects where each number is called a **term** of the sequence. Order matters: the position of each number identifies its identity!

Finite vs. Infinite

- **Infinite Sequence:** Continues indefinitely without ending (indicated by three dots ...).
  - Natural: 1, 2, 3, 4, 5, ...
  - Odd: 1, 3, 5, 7, 9, 11, ...
- **Finite Sequence:** Has a fixed, countable number of terms.
  - E.g., 6, 12, 24, 48, 96 (5 terms).

Mathematical Notation

- Subscripts connect the position to the term:
  - t_1 represents the 1st term
  - t_2 represents the 2nd term
  - t_n represents the n-th term
- For multiple sequences, we use distinct letters: t_n, s_n, u_n, v_n.

Diverse Sequences

- Terms are not limited to growing positive integers!
- **Unit Fractions (Decreasing):** 1, 1/2, 1/3, 1/4, 1/5, ...
- **Negative Integers:** –7, –3, 1, 5, 9, ... (common difference of 4).
- The index n is always natural (1, 2, 3...), but terms can be any real number.

Classroom Check: Sequences & Subscript Notation

1 What is the mathematical difference between a set of numbers {2, 5, 8} and a sequence of numbers (2, 5, 8)?

Think about whether the order of arrangement matters.

2 In the sequence 1, 3, 5, 7, 9, what are the values of t_1 and t_4?

Match each subscript with the term's position.

3 Give an everyday real-life example of a finite sequence with exactly 7 terms.

Consider days of the week or school timetable periods.

4 Can the position index n of a term t_n ever be −3 or 4.5? Can the term value itself be negative?

Recall that positions are positive counting integers.

Visual Patterns: Triangular & Square Numbers

Triangular Numbers: 1, 3, 6, 10, 15, 21, ...

Numbers represented as triangular arrays of dots (Fig. 8.1):

- Each term is the sum of consecutive natural numbers up to that rank:
  - $T_1 = 1$
  - $T_2 = 1 + 2 = \mathbf{3}$
  - $T_3 = 1 + 2 + 3 = \mathbf{6}$
  - $T_4 = 1 + 2 + 3 + 4 = \mathbf{10}$
  - $T_5 = 1 + 2 + 3 + 4 + 5 = \mathbf{15}$
- Differences between consecutive terms are: 2, 3, 4, 5, 6, ...

Square Numbers as Sums of Odd Numbers

Squares 1, 4, 9, 16, 25, 36, ... (Fig. 8.2):

- Every square number equals the sum of the first n consecutive odd numbers:
  - $1^2 = \mathbf{1}$
  - $2^2 = 1 + 3 = \mathbf{4}$
  - $3^2 = 1 + 3 + 5 = \mathbf{9}$
  - $4^2 = 1 + 3 + 5 + 7 = \mathbf{16}$
  - $5^2 = 1 + 3 + 5 + 7 + 9 = \mathbf{25}$
- Consecutive differences are the odd numbers: 3, 5, 7, 9, 11, ...

Geometric Invariant: The n-th square number is the sum of the first n odd numbers!

Classroom Check: Triangular & Square Patterns

1 What is the 6th triangular number T_6?

Add the first 6 natural numbers: 1 + 2 + 3 + 4 + 5 + 6.

2 Express the square number 36 as a sum of consecutive odd numbers.

Write down the sum of the first 6 odd numbers.

3 For the sequence 1, 4, 7, 10, 13, …, predict the next two terms and state the constant difference between them.

Observe how much is added at each step.

4 What geometric shape do you get if you combine two consecutive triangular numbers, such as T_2 (3) and T_3 (6)?

Calculate 3 + 6 and identify the special number.

Explicit Rules: Direct Calculation from Position n

DEFINITION: EXPLICIT FORMULA

An **explicit formula** calculates the value of the n-th term directly using only its position number n, without needing to know any preceding terms!

Example 1: $u_n = 2n - 1$

The sequence of odd numbers:

- $u_1 = 2(1) - 1 = 1$
- $u_2 = 2(2) - 1 = 3$
- $u_3 = 2(3) - 1 = 5$
- Find u_{53} instantly:
  $u_{53} = 2(53) - 1 = \mathbf{105}$!
- No need to list terms 1 through 52!

Testing Sequence Membership

Checking if 308 belongs to $s_n = 5n - 2$:

- Set $5n - 2 = 308$ &implies; $5n = 310$ &implies; $\mathbf{n = 62}$.
- Since 62 is a natural number, 308 is the **62nd term**.
- Now test 471:
  $5n - 2 = 471$ &implies; $5n = 473$ &implies; $\mathbf{n = 94.6}$.
- Not a natural number &implies; **471 is NOT a term!**

The Mystery of Primes

Sequence of prime numbers:

- 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, ...
- Unlike arithmetic or square sequences, **no simple explicit algebraic formula** exists that predicts every prime number!
- A reminder that mathematical patterns can be subtle and deep.

Classroom Check: Explicit Rules & Membership

1 Using the explicit rule $t_n = 3n - 7$, what is the 50th term of the sequence?

Substitute $n = 50$ into $3(50) - 7$.

2 Which position does the number 137 occupy in the odd number sequence $u_n = 2n - 1$?

Solve $2n - 1 = 137$ for n.

3 Why must the solution for n always be a positive natural number for a candidate value to be a valid term?

Consider what n represents physically.

4 What is the explicit formula for the sequence of square numbers 1, 4, 9, 16, 25, …?

Express the n-th term in terms of n.

Recursive Rules: Building Terms from Earlier Steps

DEFINITION: RECURSIVE FORMULA

A **recursive rule** defines each term by relating it to one or more preceding terms. It always requires two ingredients: an initial starting term (base case) and a step-by-step update rule!

Linear Step: $t_n = t_{n-1} + 3$

Sequence 1, 4, 7, 10, 13, ...:

- Initial term: $\mathbf{t_1 = 1}$
- Update rule: $\mathbf{t_n = t_{n-1} + 3}$ (for $n \geq 2$)
- $t_2 = t_1 + 3 = 1 + 3 = 4$
- $t_3 = t_2 + 3 = 4 + 3 = 7$
- Contrast with explicit rule: $t_n = 3n - 2$.

Non-Linear Step: $s_n = s_{n-1}(s_{n-1} - 1)$

Rapidly exploding recursion (Example 4):

- Given: $\mathbf{s_1 = 3}$
- $s_2 = s_1(s_1 - 1) = 3(2) = \mathbf{6}$
- $s_3 = s_2(s_2 - 1) = 6(5) = \mathbf{30}$
- $s_4 = s_3(s_3 - 1) = 30(29) = \mathbf{870}$
- Terms grow with dramatic speed through self-multiplication!

Classroom Check: Recursive Sequences

1 If a recursive rule is $u_1 = 1$ and $u_n = 2u_{n-1} + 3$, calculate the 2nd and 3rd terms.

Evaluate $2(1) + 3$ and then $2(5) + 3$.

2 What essential piece of information is required alongside a recursive formula before you can compute any terms?

Think about the starting point of the sequence.

3 Why is calculating the 100th term directly much harder using a recursive rule than an explicit formula?

Consider how many intermediate terms you must calculate.

4 A sequence begins at −5 and adds 3 at each step: $t_1 = -5$, $t_{n+1} = t_n + 3$. Find the first four terms.

Add 3 successively starting from −5.

The Virahānka-Fibonacci Sequence (7th Century CE)

TWO-TERM SUMMATION RULE

$V_1 = 1$, $V_2 = 2$, and $V_n = V_{n-1} + V_{n-2}$ (for $n \geq 3$) &implies; 1, 2, 3, 5, 8, 13, 21, 34, 55, ... Every term is the exact sum of the preceding two terms!

Origin in Sanskrit & Prakrit Meter

Discovered in poetic rhythm:

- Originated in ancient India while analyzing combinations of short (laghu, 1 beat) and long (guru, 2 beats) syllables in poetry!
- First formulated explicitly by **Virahānka** (c. 700 CE) in *Vṛttajātisamuccaya*.

Gopāla & Hemachandra

Continued scholarship:

- Further elaborated by linguist-mathematician **Gopāla** (c. 1135 CE).
- Systematically analyzed by the Jain polymath **Hemachandra** (c. 1150 CE), 50 years before Fibonacci!

European Transmission

Leonardo of Pisa (Fibonacci):

- Introduced to Europe in 1202 CE in *Liber Abaci* through the rabbit breeding puzzle.
- Today recognized as the **Virahānka-Fibonacci sequence**, appearing in pinecones, sunflowers, and spirals!

Classroom Check: Virahānka-Fibonacci Numbers

1 In the Virahānka sequence 1, 2, 3, 5, 8, 13, 21, 34, …, what are the next two terms?

Add 21 + 34 to find the next term, then repeat.

2 In what cultural and literary context was this sequence first discovered in ancient India?

Recall the study of short and long syllables in poetic meter.

3 Name two Indian mathematicians who studied and expanded this sequence before Fibonacci published it in 1202 CE.

Recall the scholars from the 12th century mentioned in the textbook.

4 If $T_1 = 1$, $T_2 = 2$, $T_3 = 4$, and $T_n = T_{n-1} + T_{n-2} + T_{n-3}$, what is the value of T_4?

Sum the previous three terms: 1 + 2 + 4.

Arithmetic Progression (AP): Constant Difference

DEFINITION & GENERAL FORM OF AN AP

An **Arithmetic Progression (AP)** is a sequence in which each term after the first is obtained by adding a fixed constant number **d** (common difference) to the previous term.

General Form: **a, a + d, a + 2d, a + 3d, …, a + (n – 1)d**

Growing Squares (Fig. 8.3)

Squares at stages 1, 2, 3, 4:

- Pattern: 1, 5, 9, 13, 17, 21, …
- Each stage adds 4 squares at the corners.
- First term **a = 1**
- Common difference **d = 4**
- $t_n = 1 + (n - 1) \times 4 = \mathbf{4n - 3}$

Master Formula for n-th Term

Explicit Rule:

$$t_n = a + (n - 1)d$$

- Recursive Rule:
  $\mathbf{t_1 = a}$
  $\mathbf{t_n = t_{n-1} + d}$ (for $n \geq 2$)

Decreasing APs (Negative d)

Sequence: 11, 7, 3, –1, –5, …

- First term: **a = 11**
- Common difference:
  $d = 7 - 11 = \mathbf{-4}$
- $t_n = 11 + (n - 1)(-4)$
  $= \mathbf{15 - 4n}$
- d can be positive, negative, or even fractional!

Classroom Check: Arithmetic Progression Basics

1 For the sequence 2, 5, 8, 11, …, identify the first term a, the common difference d, and write the formula for t_n.

Subtract consecutive terms: 5 − 2 = 3.

2 Why is the multiplier in the n-th term formula (n − 1) and not simply n?

Think about how many times d is added to reach the n-th term from the 1st term.

3 In the sequence 21, 18, 15, …, what is the common difference d? Is 0 a term of this AP?

Solve $21 + (n-1)(-3) = 0$ for n.

4 Is the sequence of square numbers 1, 4, 9, 16, 25, … an arithmetic progression? Why or why not?

Check whether the difference between consecutive terms is constant.

Visualising an AP: Straight Lines & Real-World Models

The Linear Graph of an AP (Fig. 8.4)

Plotting ordered pairs (Stage n, Term t_n):

- Pairs for growing squares: (1, 1), (2, 5), (3, 9), (4, 13), (5, 17)...
- **Key Discovery:** All points of an AP lie on a single, perfectly straight line!
- The slope of the line equals the common difference **d**.
- An AP is the discrete algebraic cousin of a linear equation ($y = mx + c$).

The City Taxi Fare Model (Example 5)

Charges: Fixed booking fee ₹200 + ₹40 per km travelled:

- After 1 km: ₹200 + ₹40 = **₹240**
- After 2 km: ₹200 + 2(₹40) = **₹280**
- After 3 km: ₹200 + 3(₹40) = **₹320**
- Forms an AP: $a = 240$, $d = 40$.
  $t_n = 240 + (n - 1)40 = \mathbf{200 + 40n}$
- For 10 km journey: Fare = 200 + 40(10) = **₹600**!

Classroom Check: Linear Graphs & Word Problems

1 When you plot the terms of an arithmetic progression on a coordinate plane, why do all the points fall on a straight line?

Connect this to the rate of change being constant.

2 In the taxi fare formula Fare = 200 + 40n, what does the constant 200 represent, and what does 40 represent?

Distinguish between the fixed charge and the per-km rate.

3 Harish starts work at ₹5,00,000 per year with an annual raise of ₹20,000. Write the AP formula for his salary in year n.

Identify a and d.

4 An AP has 3rd term equal to 12 and 50th term equal to 106. How can you find a and d?

Form two simultaneous equations: $a + 2d = 12$ and $a + 49d = 106$.

Sum of the First n Natural Numbers: $S_n = n(n+1)/2$

MASTER SUM FORMULA & ĀRYABHAṬA'S RULE

$S_n = 1 + 2 + 3 + \ldots + n = \mathbf{n(n+1)/2}$

First recorded in **Āryabhaṭīya (499 CE, Ch. 2, Verse 19)**: *"Half the sum of first and last terms multiplied by the number of terms."*

The Reversal-and-Pairing Method

Adding the sum forward and backward:

- $S = 1+2+3+\ldots+10$
- $S = 10+9+8+\ldots+1$
- Add column-wise: each pair sums to **11**!
  $2S = 11+11+\ldots+11$ (10 times) $= 10 \times 11 = 110$
- **$S = 110/2 = 55$**.
- For any n: $2S = n(n+1)$ &implies; **$S_n = n(n+1)/2$**!

Geometric Dot Array & Range Sums

Two triangular halves form an $n \times (n+1)$ rectangle (Fig. 8.5):

- $2 \times (1+2+3+4+5+6) = 6 \times 7 = 42$ dots &implies; Sum = 21.
- **Sum of Consecutive Ranges:**
  Calculate $25+26+\ldots+58$:
  $= S_{58} - S_{24}$
  $= (58 \times 59)/2 - (24 \times 25)/2 = 1711 - 300 =$ **1411**!

Classroom Check: Natural Number Sums

1 Using the formula $S_n = n(n+1)/2$, calculate the sum of the first 100 natural numbers: $1 + 2 + \ldots + 100$.

Compute $(100 \times 101)/2$.

2 How does the formula for the sum of natural numbers relate directly to the n-th triangular number?

Recall the definition of triangular numbers from Section 8.1.

3 A child arranges marbles in 25 rows: 1 in the first, 2 in the second, up to 25. How many marbles are used in total?

Calculate $S_{25} = (25 \times 26)/2$.

4 How would you calculate the sum of integers from 51 to 100 without adding each one individually?

Subtract S_{50} from S_{100}.

Geometric Progression (GP): Multiplicative Growth

DEFINITION & GENERAL FORM OF A GP

A **Geometric Progression (GP)** is a sequence in which each term after the first is obtained by multiplying the previous term by a fixed non-zero constant **r** (common ratio).

General Form: **$a, ar, ar^2, ar^3, \ldots, ar^{n-1}$**

Doubling Pattern (Fig. 8.6)

Green squares at stages 1, 2, 3, 4:

- Pattern: 3, 6, 12, 24, 48, 96, ...
- Each term doubles the previous count!
- First term: **a = 3**
- Common ratio:
  r = 6/3 = 12/6 = 24/12 = **2**
- Explicit rule: $t_n = 3 \times 2^{n-1}$

Explicit & Recursive Formulas

For any GP:

$$t_n = a \cdot r^{n-1}$$

- Recursive Rule:
  $t_1 = a$
  $t_n = r \cdot t_{n-1}$ (for $n \geq 2$)

Fractional & Alternating GPs

Common ratio can be any real number:

- **Shrinking GP (Example 9):**
  5, 15/4, 45/16, 135/64, ...
  a = 5, r = **3/4**.
- **Oscillating GP:**
  1, –1, 1, –1, 1, ... (r = **–1**).
- **Powers of 10:** 3, 30, 300, 3000... (r = 10).

Classroom Check: Geometric Progressions

1 For the GP 2, 10, 50, 250, …, what is the common ratio r, and what is the formula for the n-th term?

Divide 10 by 2, then write $a \cdot r^{n-1}$.

2 What is the 10th term of the geometric progression 5, 25, 125, …?

Express as a power of 5.

3 How do you test whether a given sequence is an arithmetic progression (AP) or a geometric progression (GP)?

Compare consecutive differences vs. consecutive ratios.

4 Bacteria in a culture double every hour. If there are 30 bacteria initially, how many will be present after n hours?

Model as a GP with ratio 2.

The Sierpiński Triangle: Infinite Self-Similarity

⚠ WACŁAW SIERPIŃSKI (1882–1969) & THE FRACTAL TRIANGLE

Take an equilateral triangle of area 1 (Stage 0). Join side midpoints and remove the inverted center triangle. Repeat endlessly on every remaining triangle! A fractal is a shape that repeats itself at every scale.

Number of Triangles Explodes (GP: r = 3)

Counting remaining shaded triangles:

- Stage 0: 3^0 = **1 triangle**
- Stage 1: 3^1 = **3 triangles**
- Stage 2: 3^2 = **9 triangles**
- Stage 3: 3^3 = **27 triangles**
- **Stage n: $t_n = 3^n$** (Increases extremely fast!).

Shaded Area Evaporates (GP: r = 3/4)

Tracking the total remaining black area:

- Stage 0: Area = **1**
- Stage 1: $(3/4)^1$ = **3/4 = 0.75**
- Stage 2: $(3/4)^2$ = **9/16 ≈ 0.56**
- Stage 3: $(3/4)^3$ = **27/64 ≈ 0.42**
- **Stage n: $s_n = (3/4)^n$**. As n → ∞, Area → **0**!
- *Infinite triangles with zero total area!*

Classroom Check: Sierpiński Triangle & Fractals

1 How many black triangles exist at Stage 4 and Stage 5 of the Sierpiński triangle?

Calculate 3^4 and 3^5.

2 What happens to the total shaded area of the Sierpiński triangle as the stage number n increases toward infinity?

Consider what happens when you raise 3/4 to an immense power.

3 What is a "fractal"? Name two examples of fractal-like structures found in biological nature.

Recall the vegetable and weather examples mentioned in the textbook.

4 What are the recursive rules for the triangle count t_n and the shaded area s_n of the Sierpiński triangle?

Express t_n in terms of t_{n-1} and s_n in terms of s_{n-1}.

Visualising a GP: Non-Linear Curves & Bouncing Balls

Curved Graphs of a GP (Fig. 8.9 & 8.10)

Plotting ordered pairs (Stage n, Term t_n):

- Pairs for GP 3, 6, 12, 24, 48: (1, 3), (2, 6), (3, 12), (4, 24), (5, 48).
- **Fundamental Contrast with AP:** Points of a GP do **NOT** lie on a straight line!
- When $r > 1$, points trace a rapidly steepening upward curve (exponential growth).
- When $0 < r < 1$, points trace a smoothly decaying curve flattening toward the x-axis.

The Bouncing Ball Model (Example 10)

Dropped from 24 feet; rebounds to 3/4 (75%) of previous height:

- Initial: 24 ft | Bounce 1: **18.0 ft**
- Bounce 2: **13.5 ft** | Bounce 3: **10.125 ft**
- Bounce 4: **7.59 ft** | Bounce 5: **5.70 ft**
- Bounce 6: **4.27 ft** | Bounce 7: **3.20 ft**
- 1/6th of initial height = 24 / 6 = **4.0 ft**.
  After **7 bounces**, the rebound height drops below 4 ft!

Classroom Check: Bouncing Physics & Curves

1 Why do the plotted points of an AP form a straight line, whereas the plotted points of a GP form a curved graph?

Contrast constant addition with constant multiplication.

2 A ball dropped from 80 m bounces back to 60% of its previous height each time. What height does it reach on the 1st and 2nd bounces?

Calculate 80×0.60 and then 48×0.60.

3 In the 24 ft bouncing ball problem, what is the first term a and common ratio r of the rebound heights sequence?

Note that the rebound sequence begins after the 1st bounce.

4 Which term of the geometric progression 2, 6, 18, 54, … is equal to 4374?

Solve $2 \times 3^{n-1} = 4374$.

The Sierpiński Square Carpet: Trisection & Holes

CONSTRUCTION OF THE SIERPIŃSKI CARPET (FIG. 8.12)

Start with a unit square (Stage 0). Trisect each side into 3 equal parts to form 9 smaller squares. Remove the central square, retaining 8 smaller squares. Repeat endlessly on every retained square!

Number of Retained Squares (GP: r = 8)

Powers of 8 across stages:

- Stage 0: $8^0 =$ **1 square**
- Stage 1: $8^1 =$ **8 squares**
- Stage 2: $8^2 =$ **64 squares**
- Stage 3: $8^3 =$ **512 squares**
- Stage 4: $8^4 =$ **4096 squares**
- **Stage n: $t_n = 8^n$** (Recursive: $t_n = 8 \cdot t_{n-1}$).

Area of Shaded Carpet (GP: r = 8/9)

Tracking the total remaining area:

- Stage 0: Area = **1**
- Stage 1: $(8/9)^1 =$ **$8/9 \approx 0.889$**
- Stage 2: $(8/9)^2 =$ **$64/81 \approx 0.790$**
- Stage 3: $(8/9)^3 =$ **$512/729 \approx 0.702$**
- **Stage n: $s_n = (8/9)^n$**
- As stages continue forever, Area → **0** while the boundary length explodes toward infinity!

Classroom Check: Sierpiński Carpet

1 Into how many small squares is the unit square divided at Stage 1 of the Sierpiński carpet, and how many are removed?

Trisect each dimension: $3 \times 3 = 9$.

2 What is the common ratio r for the sequence representing the number of shaded squares in the carpet?

Count how many smaller squares replace each square at the next stage.

3 Why is the common ratio for the remaining area of the carpet equal to 8/9?

Think about the fraction of area retained at each trisection.

4 Write down the explicit formula for both the square count t_n and the remaining area s_n at stage n of the carpet.

State both formulas in terms of powers of n.

Model Problem Solutions from Exercises

Problem 1: AP Terms (Ex 8.2 Q4)

Given: AP has 50 terms; 3rd term = 12, last term (t_{50}) = 106. Find t_{29}.

- $a + 2d = 12$
- $a + 49d = 106$
- Subtract: $47d = 94$ &implies; **$d = 2$**
- $a = 12 - 2(2) =$ **8**
- $t_{29} = 8 + (28)(2) = 8 + 56 =$ **64**!

Problem 2: 12th Term of GP (Ex 8.3 Q1)

Given: GP with common ratio $r = 2$, whose 8th term is 192. Find t_{12}.

- $t_8 = a \cdot r^7 = 192$
- $a \cdot 2^7 = 192$ &implies; $a \cdot 128 = 192$
- $a = 192 / 128 =$ **3/2**
- $t_{12} = t_8 \cdot r^4 = 192 \times 2^4$
  $= 192 \times 16 =$ **3072**!

Problem 3: Natural Sum > 1000 (Ch Ex Q9)

Given: Find the smallest n such that $S_n = n(n+1)/2 > 1000$.

- $n(n+1) > 2000$
- Approximate $\sqrt{2000} \approx 44.7$
- Test $n = 44$:
  $(44 \times 45)/2 = 22 \times 45 =$ **990** (≤ 1000)
- Test $n = 45$:
  $(45 \times 46)/2 = 45 \times 23 =$ **1035** (> 1000)
- **Smallest n = 45**!

Master Sequences & Progressions Checklist

Sequences & Arithmetic Progressions

- ✅ **Sequence Definition:** Ordered list of numbers where position matters; finite or infinite.
- ✅ **Explicit Formula:** Direct calculation of t_n from position n without earlier terms.
- ✅ **Recursive Formula:** Updates terms based on predecessor(s): e.g., $t_n = t_{n-1} + d$.
- ✅ **Virahānka–Fibonacci:** $V_1=1$, $V_2=2$, $V_n=V_{n-1}+V_{n-2}$ (Virahānka 700 CE; Gopāla & Hemachandra).
- ✅ **Arithmetic Progression (AP):** Constant difference d: $\mathbf{t_n = a + (n-1)d}$.
- ✅ **AP Graph:** Points (n, t_n) always lie on a single straight line with slope d.
- ✅ **Sum of Natural Numbers:** $S_n = \mathbf{n(n+1)/2}$ (Āryabhaṭīya 499 CE).

Geometric Progressions & Fractals

- ✅ **Geometric Progression (GP):** Constant multiplier r: $\mathbf{t_n = a \cdot r^{n-1}}$.
- ✅ **Recursive Rule for GP:** $t_1 = a$, $t_n = r \cdot t_{n-1}$ (for $n \geq 2$).
- ✅ **GP Graph:** Points (n, t_n) trace a non-linear exponential curve, never a straight line!
- ✅ **Fractal Concept:** Infinite geometric self-similarity repeating across shrinking scales.
- ✅ **Sierpiński Triangle:** Triangle count $t_n = 3^n$ (GP: $r = 3$); Area $s_n = (3/4)^n$ (GP: $r = 3/4 \to 0$).
- ✅ **Sierpiński Carpet:** Square count $t_n = 8^n$ (GP: $r = 8$); Area $s_n = (8/9)^n$ (GP: $r = 8/9 \to 0$).
- ✅ **Bouncing Ball Physics:** Height after successive bounces follows a decaying GP ($r < 1$).

END OF CHAPTER 8

THANK YOU

"Mathematics is the science of patterns, and the art of predicting what comes next — turning the rhythm of poetry and the growth of nature into universal laws of reason."

✦ Akash Srivastva (www.akashmaths.online)